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Predictive modeling and anomaly detection in solar PV inverters using machine learning

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ABSTRACT

The operational stability of photovoltaic (PV) systems is critical to the success of distributed renewable energy integration. This study presents a machine learning-driven framework for performance modeling, anomaly detection, and classification of inverter output in a grid-connected PV installation. Using high-resolution data collected from 30 kW and 40 kW inverters over one month, we applied supervised learning techniques to predict active power output, categorize production levels, and detect deviations from expected behavior. A Random Forest Regressor achieved high fidelity in power prediction ($R^2 = 0.995$, MAE $= 0.12$ kW), while classification models categorized output levels with 100 % accuracy under static conditions. Anomaly detection using Z-score analysis identified significant outliers, particularly during high-production intervals. However, one-hour-ahead classification revealed substantial drops in predictive performance (accuracy = 36.4 %), highlighting the inherent difficulty of forecasting under variable environmental conditions. Feature importance analysis underscored the dominant role of inverter input power and phase currents in prediction accuracy. These findings demonstrate the utility of interpretable, data-driven models for real-time diagnostics and forecasting in PV systems and lay the groundwork for scalable smart monitoring infrastructures. Unlike prior studies that rely on meteorological inputs, this work uses only inverter and grid-side electrical measurements, demonstrating that interpretable models can provide actionable insights even in data-limited scenarios. The proposed framework offers a practical foundation for early-warning diagnostics and intelligent maintenance scheduling, enabling more resilient PV operations. Its integration of interpretable machine learning with real-time monitoring distinguishes it from conventional rule-based supervision methods.

1. Introduction

Global decarbonization strategies place wind and solar power at the center of the energy transition. According to the International Energy Agency (IEA), reaching net-zero emissions by 2050 will depend on these two sources providing roughly 70 % of total electricity generation. Reflecting this ambition, the European Union aims for an 80 % reduction in greenhouse gas emissions from 1990 levels and a complete shift to renewable power by 2050 [1]. Driven by these global decarbonization efforts, the deployment of photovoltaic (PV) systems has expanded rapidly in recent years. However, maintaining optimal performance and operational reliability of grid-connected PV plants remains a persistent challenge due to fluctuating weather conditions, system faults, and grid disturbances [2]. These factors can cause substantial performance and financial losses, with inverter-related faults reported to reduce annual energy yield by 10–70 % and increase maintenance costs by up to 20 % [3]. Inverters are particularly critical with regards to PV system reliability as they typically fail a number of times during the overall system lifetime [4–8]. Consequently, accurate monitoring, forecasting, and classification of inverter behavior are essential for maximizing energy yield and minimizing operational downtime

In recent years, the integration of data-driven approaches into energy systems has opened new opportunities for intelligent diagnostics and forecasting. Machine learning (ML) techniques, in particular, have demonstrated strong capabilities in modeling complex and nonlinear relationships within energy production data [9–11]. Despite these advances, many existing methods—especially deep learning architectures [12], suffer from low interpretability, high computational demands, and

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reliance on large labeled datasets or environmental inputs such as irradiance and temperature [13]

This study addresses these gaps by developing and interpretable and computationally lightweight framework suitable for real-time PV inverter monitoring even when environmental sensors are unavailable. This paper presents a machine learning-based framework for analyzing and classifying inverter performance in a commercial PV installation using electrical and temporal data alone. Using high-resolution operational data (5-minute intervals) collected from 30 kW and 40 kW inverters, as well as grid voltage and power sensors, we investigate the following research questions:

1. Can machine learning models accurately classify inverter production levels across multiple output classes (e.g. low, normal, high) under real-time conditions?
2. To what extent these models forecast inverter performance categories one hour in advance, given variable environmental dynamics?
3. Which input features contribute most to predictive accuracy and system interpretability?
4. How can residual-based statistical analysis (e.g., Z-score) enhance detection of production anomalies?

The chosen algorithms, Random Forests for prediction and Z-score analysis for anomaly detection, were selected for their robustness, interpretability, and suitability for small yet high-frequency datasets, making them well-aligned with practical PV monitoring deployments. Furthermore, the absence of irradiance or temperature data (a common limitation in real-world monitoring) is explicitly addressed by constructing time-based proxies (hour, day, and weekday patterns) to capture cyclical solar generation behavior.

Random Forest algorithms have been widely adopted in PV system modeling; however, previous studies generally focus on a single analytical task, such as power forecasting or fault classification, and frequently depend on aggregated datasets supplemented with meteorological information [14,15]. This study advances the section Related work by presenting a unified, data-driven framework that simultaneously performs regression, classification, and anomaly detection using only raw inverter and grid-side data—eliminating the need for irradiance and temperature measurements [16]. This design allows for comprehensive performance assessment, fault sensitivity analysis, and operational interpretability within a single, scalable workflow. Furthermore, the adoption of time-aware cross-validation and detailed feature importance analysis provides novel insights into inverter–grid interactions and model generalization under real operating conditions. These aspects collectively represent a distinct methodological contribution beyond prior Random Forest-based PV studies. By combining time-series preprocessing, supervised learning, and statistical anomaly detection techniques, this study demonstrates the potential for scalable and interpretable performance monitoring and forecasting in PV systems. Our results show that production class can be predicted with high accuracy, and that grid-side variables such as reactive power and voltage imbalances are highly informative. The proposed approach provides not only accurate short-term modeling, but also actionable insights for maintenance scheduling and operational optimization and could serve as a foundation for smarter energy management in distributed solar power plants.

2. Related work

The growing complexity of photovoltaic (PV) installations has intensified the need for reliable, data-driven approaches to performance monitoring and fault detection. Inverter-level monitoring, in particular, plays a central role in ensuring grid stability and operational reliability. Recent advances in machine learning (ML) have enabled more accurate diagnostics and forecasting; however, many approaches remain computationally heavy, rely on large labeled datasets, or depend on environmental measurements that are not always available in real-world PV systems. This literature review synthesizes current methodologies for PV anomaly detection, examining various methods as machine learning approaches, statistical methods, signal processing techniques, and hybrid solutions developed between 2020–2025 years.

2.1. Statistical and signal-based fault detection methods

Early approaches to photovoltaic (PV) system monitoring relied on statistical indicators and signal processing techniques to detect operational anomalies. Statistical control charts such as the Exponentially Weighted Moving Average (EWMA) method have been successfully applied for residual-based fault detection with low computational overhead [17]. Similarly, hypothesis testing approaches like the Generalized Likelihood Ratio Test (GLRT) enable early identification of abnormal behavior based on machine-learning-computed residuals [18].

Signal processing methods have complemented statistical models by extracting characteristic features from inverter or string-level signals. Multi-resolution signal decomposition combined with fuzzy inference systems provides robust fault detection under low-irradiance conditions [19]. Likewise, the Stockwell transform has been used for time–frequency analysis, enabling highly accurate fault classification when integrated with data mining algorithms [20]. Wavelet-based techniques also contribute to islanding detection and inverter protection through frequency-domain feature extraction [21].

Although these methods are computationally efficient and interpretable, they depend heavily on predefined thresholds and domain heuristics, limiting their scalability and adaptability under dynamic environmental conditions.

2.2. Machine learning approaches for PV system diagnostics

The adoption of machine learning (ML) methods has enabled data-driven fault classification and performance modeling in PV systems. Various supervised algorithms have been tested to detect anomalies directly from inverter or SCADA data. Random Forest (RF), k-nearest neighbors (k-NN), and Linear Discriminant Analysis (LDA) models have demonstrated accuracies between 94 % and 100 % across different PV installations [22,23].

Support Vector Machines (SVM) remain among the most widely applied techniques, offering strong classification performance, particularly when optimized with metaheuristics. The combination of SVM and Grey Wolf Optimization, for instance, enhances the identification of anomalies in large-scale solar power plants [24], while binary salp swarm optimization improves fault diagnosis from three-phase current data [25].

Studies comparing ML algorithms reveal consistent results: tree-based ensembles and SVMs outperform purely statistical baselines in both accuracy and robustness [26,27]. However, most of these approaches still depend on large labeled datasets, and their ability to generalize across sites remains limited.

2.3. Deep learning and hybrid architectures

Recent advances in deep learning have led to the development of highly accurate yet computationally demanding frameworks for PV anomaly detection. Autoencoders and LSTM-based hybrids (AE-LSTM) have achieved strong performance in identifying abnormal inverter behavior using only power output signals [28,29]. Convolutional Neural Networks (CNNs), often combined with image-based transformations such as Gramian Angular Fields, have been successfully used for grid-connected inverter fault detection, reaching accuracy levels above 95 % [30,31].

Transformer-based architectures have recently emerged as a frontier for time-dependent PV analysis. The anomaly-transformer technique, for example, provides superior fault localization under both sunny and cloudy conditions [32]. Likewise, the Multiscale Linear Attention and Scale Distribution Alignment Learning (MLA-SDAL) framework shows promising results in electroluminescence-based anomaly identification [33].

Hybrid architectures that combine fuzzy systems, regression trees, neural networks, and Gaussian Process Regression (GPR) have demonstrated exceptional accuracy for PV string-level fault detection [34]. These models often achieve mean absolute errors below 0.01 kW, highlighting the predictive potential of multi-method integration.

Despite their accuracy, deep and hybrid models face key challenges: interpretability, high computational cost, and dependence on extensive data preprocessing or sensor inputs such as irradiance and temperature.

2.4. Interpretable and ensemble models

To balance accuracy and interpretability, ensemble-based models have gained increasing attention. Studies indicate that ensemble algorithms such as Random Forest and Extra Trees consistently rank among the top-performing methods for PV fault detection and energy prediction [35,36]. The Extra Trees classifier, for instance, achieved perfect accuracy for fault detection and 94.4 % for classification tasks [23].

These models are particularly appealing for operational monitoring because they handle mixed data types, mitigate overfitting, and remain computationally feasible for real-time applications. Furthermore, their structure allows for direct feature importance analysis, enhancing explainability—a critical requirement for industrial diagnostics.

Hidden Markov Models (HMM) and Isolation Forests also contribute to interpretable anomaly detection. HMMs model state transitions across PV sites with median precision above 80 %, providing statistical insight into fault persistence [37]. Isolation Forests, by contrast, detect anomalies without labeled data, making them suitable for unsupervised monitoring where ground-truth faults are unavailable [38,39].

The growing integration of artificial intelligence and Internet of Things (AIoT) technologies further extends interpretability by enabling synchronized, real-time analytics and visual monitoring dashboards [40].

2.5. Limitations and research gap

While existing studies demonstrate remarkable progress, several open challenges remain. Many deep and hybrid models lack transparency, making it difficult for operators to interpret detection outcomes and act upon them [36]. Most supervised approaches require

Table 1
Summary of recent approaches for anomaly detection and performance monitoring in PV systems (2020–2025).

Study / Year	Methodology	Data source / features	Main findings	Limitations
[17] / (2017)	EWMA statistical control charts	Residuals from one-diode model	Low-cost, real-time fault detection	Limited adaptability to non-stationary data
[19] / (2017)	Multi-resolution signal decomposition + fuzzy inference	DC-side current and voltage	Robust under low irradiance	Requires manual feature extraction
[18] / (2019)	GLRT + ML residual analysis	Simulated PV fault data	Effective hypothesis-based detection	Needs statistical model assumptions
[30] / (2021)	CNN (modified AlexNet)	Grid-connected inverter data	99 % classification accuracy	Computationally demanding
[27] / (2021)	ML + IoT fault diagnosis	Field PV array data	Practical online diagnosis	Dependence on external sensors
[38] / (2021)	Isolation Forest	SCADA time-series	Detects unlabeled anomalies	Sensitive to dataset characteristics
[28] / (2022)	AE-LSTM hybrid	Power production time series	Distinguishes healthy/abnormal states	Requires long sequences, low interpretability
[37] / (2022)	Hidden Markov Model	Multi-site PV datasets	82 % median precision, interpretable states	Limited spatial transferability
[26] / (2023)	SVM vs. statistical comparison	Real PV plant data	ML outperforms statistics by 26 %	Needs labeled datasets
[39] / (2023)	DBSCAN, k-means, Isolation Forest	Unlabeled module data	Good unsupervised detection	Sensitive to hyperparameters
[24] / (2023)	SVM + Grey Wolf Optimization	PV power features	Optimized fault identification	High tuning complexity
[23] / (2023)	Extra Trees Classifier	Array electrical data	100 % detection accuracy	Overfitting risk, small dataset
[29] / (2023)	Deep Autoencoder	PV power output only	82 % F1-score	Low transparency
[33] / (2024)	MLA-SDAL + attention mechanism	Electroluminescence images	Accurate cell anomaly detection	Needs imaging sensors
[31] / (2024)	CNN + Gramian Angular Field	PV inverter signals	95.8 % accuracy	High computational cost
[25] / (2024)	Optimized SVM + binary salp swarm	Three-phase current data	Effective current-based diagnosis	Limited scalability
[20] / (2024)	Stockwell transform + data mining	PV array electrical data	99.6 % accuracy	Complex preprocessing
[32] / (2024)	Anomaly Transformer	Time-series PV data	F1 $>$ 0.97 (cloudy)	Requires large training sets
[36] / (2024)	Ensemble (RF, Extra Trees, SVM)	Real PV datasets	AUC $\approx$ 0.98	Moderate interpretability
[35] / (2024)	Sequential ensemble of RF/ET	Current + voltage characteristics	Improved temporal assessment	Computational overhead
[40] / (2024)	AIoT dynamic detection	Synchronized IoT sensors	92.6 % max accuracy	Hardware-intensive
[22] / (2025)	ML-based simulation study	Synthetic PV faults	$>$94 % detection rate	Lack of real-world validation
[34] / (2020)	Hybrid (fuzzy + ML models)	PV string DC energy	MAE = 0.002 kW	High model complexity

labeled data and often depend on environmental inputs such as irradiance and temperature [32,38], which are not always available in real-world PV installations.

Moreover, environmental variability and dataset imbalance limit the generalization of trained models across different sites and operating conditions [36]. Statistical methods, though interpretable, often miss contextual information, while deep networks sacrifice interpretability for accuracy.

To bridge these gaps, recent research calls for models that are both interpretable and computationally lightweight—capable of performing in real time with limited sensor inputs. In this context, ensemble-based frameworks such as Random Forest, combined with statistical anomaly indicators like the Z-score, offer a compelling balance between transparency, scalability, and predictive power. These methods provide the foundation for practical, data-efficient PV system monitoring, aligning with the objectives of the present study.

As shown in Table 1, the more studies achieve excellent detection accuracy but often rely on environmental or image data, complex architectures, or extensive computation. Only a few works employ interpretable ensemble models that can operate using electrical signals alone. This gap motivates the present study's focus on a lightweight Random Forest + Z-score framework that preserves transparency while ensuring reliable, real-time anomaly detection for PV inverters without environmental sensors.

Based on the research of relevant academic literature, it can be concluded that the field of photovoltaic system anomaly detection continues to develop rapidly. New research is constantly emerging [41] with hybrid approaches [42,43] and advanced artificial intelligence techniques showing the most promising solutions to current limitations while maintaining high performance standards [44].

3. Materials and methods

This study utilizes real-world operational data collected from a grid-connected photovoltaic (PV) plant located in Slovakia. The system includes two inverters with rated capacities of 30 kW and 40 kW respectively, along with grid monitoring sensors that capture voltage and power metrics across three phases. The data was exported via the Huawei FusionSolar platform and spans the period from January 1st to February 1st, 2025.

Data sources

- Inverter Data: Active power output (kW) from the 30 kW and 40 kW inverters at 5-minute resolution.
- Grid Power Data: Active and reactive power (W, var) for each grid phase.
- Grid Voltage Data: Voltage readings for phases A, B, and C.
- Environmental Context: While direct irradiance data were unavailable, we derived time-based features (hour, day, and weekday/weekend) to serve as indirect procies for solar exposure and diurnal cycles. This substitution was necessary due to the absence of on-site meteorological sensors. Althoug such proxies cannot capture short-term weather fluctuations (e.g., cloud transients), they preserve the dominant daily production patterns and allow consistent feature generation across sites lacking environmental instrumentation.

The analyzed PV installation is located in western Slovakia, representing a temperate Central European climate with typical mid-latitude solar irradiance variability. Each of these data streams was processed and aligned into a single, time-indexed dataset using *pandas*. Missing data, mostly during nighttime hours when no generation occurs, were removed when affecting label quality.

Preprocessing and Feature Engineering

To enable machine learning analysis, the following preprocessing steps were performed:

- Timestamp alignment: All data was resampled and merged at a consistent 5-minute frequency.
- Feature normalization and transformation: Selected features were scaled or derived from existing ones, including:
  - hour of the day
  - *dayofweek* and *is_weekend* flag
  - *time_index* (cumulative time progression)
- Label construction:
  - For regression, the target was continuous active power output from the inverters.
  - For classification, inverter output was discretized into three operational classes namely Low, Medium, and High by using quantile thresholds at the 33rd and 66th percentiles. This quantile-based approach ensures balanced class representation, which facilitates stable model training. However, these thresholds do not necessarily correspond to absolute inverter capacity limits or grid-defined operational states. They are intended primarily as data-driven categories for comparative modeling. Future work could incorporate domain-informed thresholds (e.g., based on inverter nominal power ranges or grid efficiency standards) to further align classification outcomes with operational interpretation. Unsupervised clustering methods (e.g., K-Means or Gaussian Mixture Models) could also be explored to derive data-driven class boundaries aligned with natural production regimes.
  - For forecasting, the labels were shifted by 12 time steps (i.e., 1 h ahead) to create predictive tasks.

For one-hour-ahead forecasting, labels were generated by shifting the target variable by 12 time steps (corresponding to 5-minute intervals). To preserve temporal dependencies without inflating feature dimensionality, the model input consisted of contemporaneous electrical and time-based features rather than explicit lag variables. This approach captures diurnal and cyclical patterns through the hour, day, and weekday indicators, while avoiding potential information leakage across training and test splits. While this statistical method simplifies label creation and supports robust classifier convergence, it does not necessarily align with physically meaningful thresholds such as inverter rated power levels or grid operating zones. A more domain-informed discretization (e.g., defining production classes based on inverter specifications or standardized power quality bands) could yield categories with clearer operational interpretation. Future work could therefore explore hybrid binning strategies that balance statistical uniformity with physical relevance and could incorporate explicit temporal features such as rolling averages or lagged power values to better represent short-term autocorrelation in inverter output.

The study explores multiple supervised machine learning techniques to assess inverter behavior and forecast performance.

Regression Modeling

Random Forest (RF) models were selected for both regression and classification tasks due to their proven robustness, interpretability, and suitability for structured, tabular datasets with limited sample size and mixed feature types. Unlike sequential deep learning models such as Long Short-Term Memory (LSTM) networks, which require large volumes of continuous temporal data and extensive hyperparameter tuning, RF models can effectively capture nonlinear dependencies without explicit temporal ordering. Gradient-boosting methods such as XGBoost or LightGBM were also considered; however, preliminary tests indicated only marginal performance gains (<1 % R^2 improvement) at the cost of reduced transparency and higher computational demand. Support Vector Machines (SVM) were excluded primarily due to scalability limitations and sensitivity to feature scaling when applied to multivariate energy datasets. The choice of RF therefore reflects a balance between predictive accuracy, interpretability, and deployment simplicity. The study prioritized models that can be readily integrated into practical PV monitoring environments.

RandomForestRegressor was trained to predict the instantaneous power output of the inverters based on grid-side features and temporal indicators. Model performance was evaluated using:

- R^2 score
- Mean Absolute Error (MAE)

Multi-Class Classification

To capture operational states, inverter output was categorized into three performance bands. A *RandomForestClassifier* was trained on grid and time-based features to classify:

- Current inverter state
- Inverter state 1 h into the future

Performance was assessed using:

- Accuracy
- F1-score
- Confusion matrix visualization

For the one-hour-ahead task, we used 5-fold *TimeSeriesSplit* to preserve chronological order and avoid temporal leakage. We report two balancing strategies: (i) *class_weight="balanced"* in the Random Forest (no synthetic data), and (ii) SMOTE applied only within each training fold, followed by evaluation on the untouched validation slice. Out-of-fold (OOF) predictions were aggregated over the evaluated region (earliest indices excluded by design in *TimeSeriesSplit*), and we summarize accuracy, macro F1, and balanced accuracy.

Anomaly Detection

We applied a Z-score-based anomaly detection technique on the inverter power output:

- A rolling mean was computed from historical predictions
- Z-scores were calculated to flag any deviations beyond ±3 standard deviations
- Anomalous periods were cross-validated against time-of-day and grid behavior

We validated the Z-score threshold by scanning $z \in [2.0, 4.25]$ in 0.25 increments and recording the fraction of samples flagged as anomalies. Across the inverter-40 kW series, the share flagged decreased monotonically from 6.42 % at $z = 2.0$ and to 0.18 % at $z = 4.25$. At the conventional $z = 3.0$ setting, 2.77 % of samples were flagged. This rate is substantially higher than the 0.27 % expected under a standard normal assumption, indicating heavier-tailed residuals and justifying an empirical sweep rather than relying solely on Gaussian assumptions. In subsequent analyses we report results at $z = 3.0$ as a pragmatic balance between sensitivity and alert volume and include a sensitivity range for context.

Visualization and Exploratory Analysis

To understand system behavior and support feature engineering, several visualizations were used:

- Hourly production profiles for both inverters
- Missing data heatmaps
- Power and voltage correlation matrices
- Feature importance rankings (via model internals and SHAP)

All analyses were performed using Python in Google Colab, with libraries including pandas, scikit-learn, matplotlib, seaborn, and shap. The overall methodology can be seen on the flow chart (Fig. 1) below.

This study focuses on Random Forest models and Z-score analysis due to their interpretability, robustness, and suitability for small to medium-sized datasets. While alternative machine-learning methods such as Support Vector Machines, XGBoost, or LSTM networks could also be applied to PV performance modeling, the goal here was to develop a transparent, data-efficient baseline framework rather than to optimize absolute predictive accuracy. Future work will include comparative benchmarking across these architectures to evaluate trade-offs between interpretability, computational cost, and temporal learning capacity.

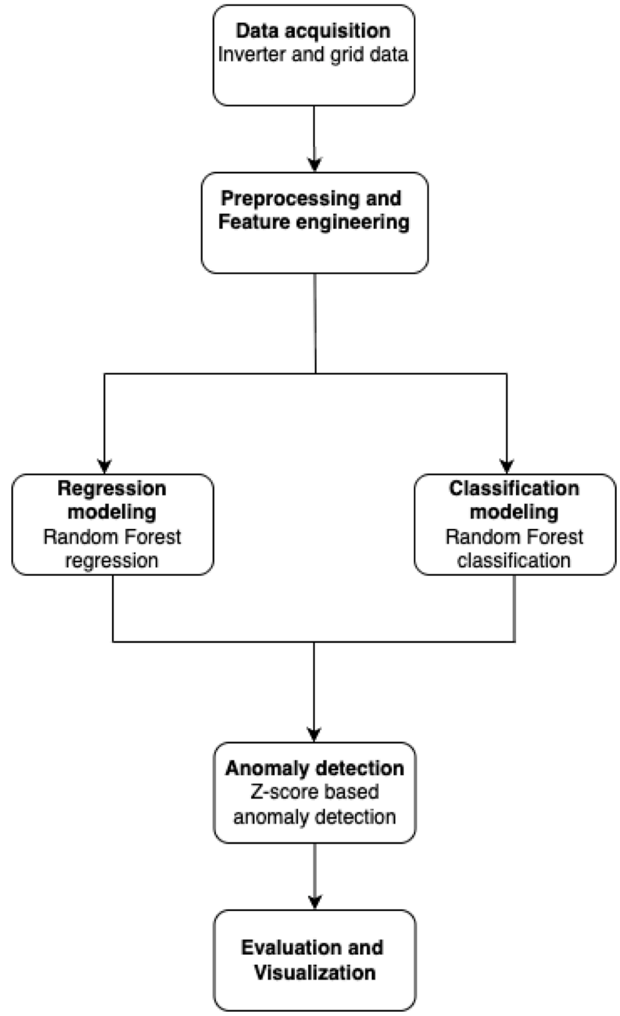

Fig. 1. Overview of the methodology for predictive modeling and anomaly detection in photovoltaic (PV) systems. The workflow includes (1) data acquisition, (2) preprocessing and feature engineering, (3) Random Forest-based regression and classification modeling, (4) Z-score-based anomaly detection, and (5) evaluation and visualization.

Similarly, for anomaly detection, the present work employed a Z-score approach to highlight statistically significant deviations in inverter output. Although simple, this method provides intuitive and easily interpretable results for operational diagnostics. In subsequent studies, we plan to benchmark this against multivariate or unsupervised techniques (e.g., Isolation Forests, Autoencoders, and statistical hypothesis testing) to assess whether they capture contextual or system-level anomalies more effectively.

4. Results and discussion

To establish a baseline understanding of inverter behavior, we analyzed the average daily power output of the 30 kW and 40 kW inverters across all days in the dataset. Fig. 2 illustrates the mean hourly production profile computed from 5-minute resolution data between January and February 2025.

As seen in the figure, both inverters exhibit a classic bell-shaped solar production curve:

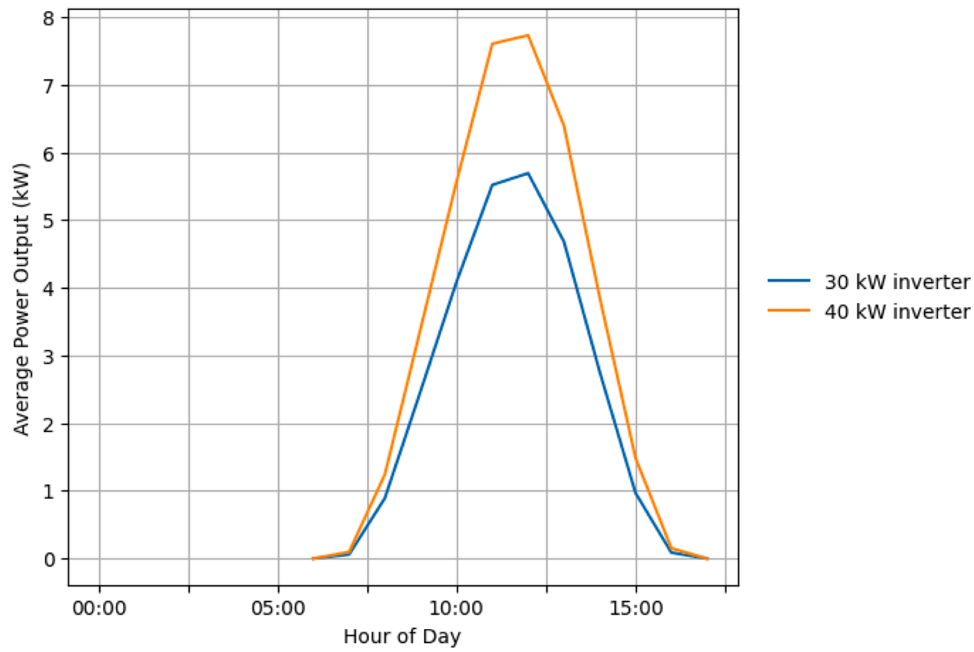

Fig. 2. Average Daily Production Profile. The x-axis represents the hour of the day, and the y-axis shows average output in kilowatts.

- Power generation begins around 07:00, gradually ramps up, and peaks between 12:00 and 13:00, corresponding to solar noon.
- The 40 kW inverter (orange line) consistently produces more energy than the 30 kW unit, as expected based on rated capacity.
- After 13:00, output declines symmetrically until complete shutdown around 17:00–18:00, consistent with winter daylight hours in Slovakia.

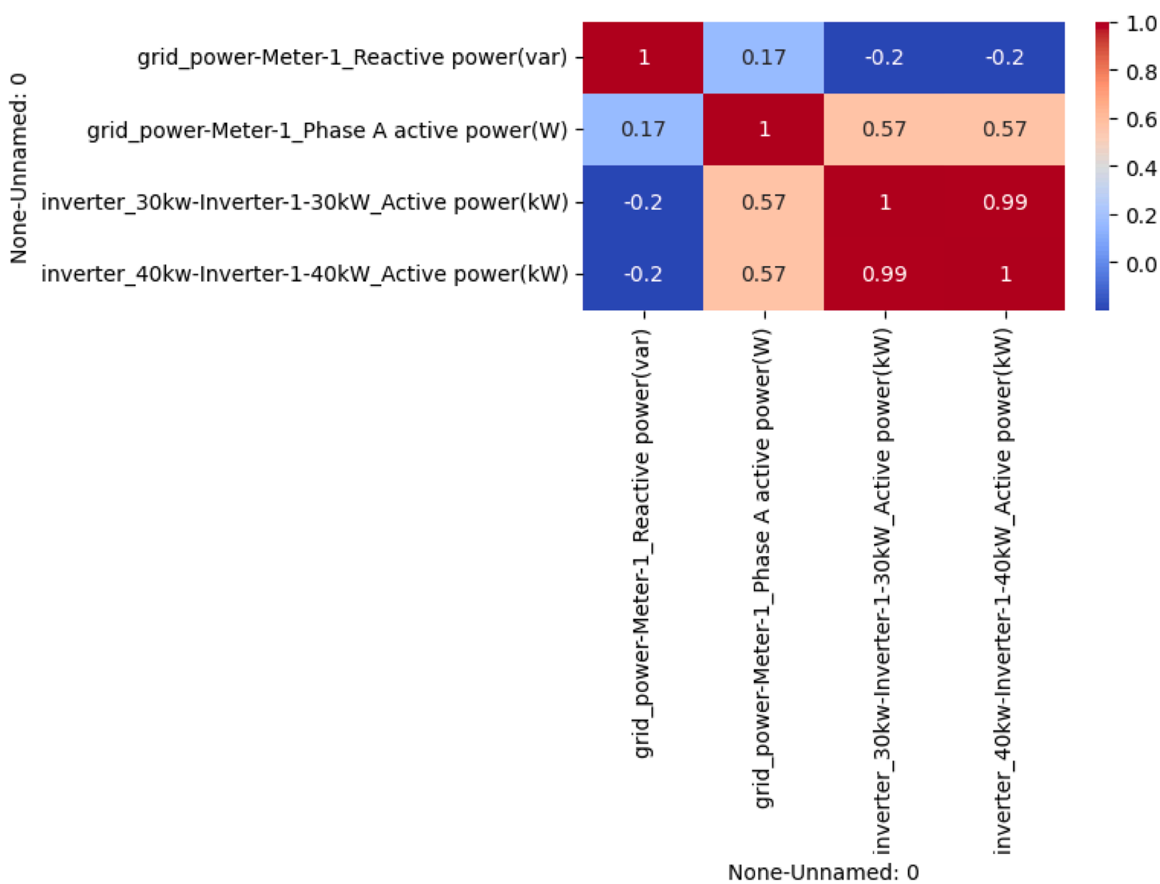

Fig. 3. Correlation Matrix of the Electrical Variables.

- The consistent shape and timing of the curves suggest stable inverter behavior, low shading, and limited fault occurrences during most days.

This profile served as the reference distribution for detecting anomalies and establishing performance categories in subsequent modeling.

Fig. 3 presents a correlation matrix evaluating the relationships among key electrical variables measured at the solar plant, including active power outputs from both inverters (30 kW and 40 kW), Phase A active power from the grid meter, and grid reactive power. The correlation coefficients were computed using Pearson's method over the full dataset duration.

A notably strong positive correlation (r= 0.99) was observed between the active power output of the 30 kW and 40 kW inverters. This result confirms that the inverters exhibit nearly identical production patterns, likely driven by shared environmental conditions such as solar irradiance and temperature. This consistency suggests the potential to use one inverter's data to validate or even predict the performance of the other, supporting redundancy analysis and model generalization strategies.

Moderate correlations (approximately r= 0.57) were identified between each inverter's active power output and the grid's Phase A active power. This indicates a reasonably direct relationship between inverter production and grid injection, though the correlation is not perfect. This gap implies that grid active power is also influenced by other system dynamics, including local consumption and losses, which are not captured solely by inverter outputs.

In contrast, the grid's reactive power demonstrated low to slightly negative correlations with all other variables (ranging around −0.20). This outcome suggests that reactive power behaves independently of active power generation and is likely governed by power factor correction mechanisms or varying inductive and capacitive loads. Such decoupling reflects the inverter's adherence to grid-code requirements for reactive power support and voltage regulation rather than energy conversion itself.

From a grid-integration perspective, these relationships highlight how fluctuations on the grid side (particularly in reactive power and phase voltage) can influence inverter operating conditions and efficiency. Transient voltage imbalances or poor power factor at the point of common coupling may induce additional stress on inverter control loops, potentially affecting long-term reliability. Conversely, stable voltage profiles support optimal inverter operation and smoother grid injection.

Therefore, understanding these interdependencies is essential for designing coordinated inverter-grid control strategies that enhance power-quality stability, minimize harmonic distortion, and improve the overall resilience of distributed PV networks.

Overall, this analysis highlights the reliability of inverter output patterns, the partial influence on grid behavior, and the decoupled nature of reactive power. These findings inform the choice of features for predictive modeling and the necessity of treating certain power characteristics as distinct phenomena within the system.

Fig. 4 illustrates the Z-score analysis performed on the active power output of the 40 kW inverter across the one-month observation period. Z-scores were calculated to detect statistical anomalies, defined here as deviations exceeding ±3 standard deviations from the mean.

The results show several spikes in Z-score values that surpass the +3 threshold (marked by the red dashed line), indicating significant outliers in the inverter's output profile. These anomalies are not randomly distributed but instead appear in clusters notably around January 10 and again from January 24–31. Such temporal concentration suggests recurring or sustained deviations rather than isolated sensor noise or random fluctuations.

These anomalous readings could be attributed to transient environmental conditions (e.g., cloud flicker, snow cover), operational disruptions (e.g., inverter shutdowns or power curtailment), or data acquisition errors. Importantly, no comparable negative outliers below the −3 threshold were observed, reinforcing that the deviations are primarily characterized by unexpected surges rather than drops.

The detection of these anomalies is crucial for operational reliability and maintenance scheduling. It allows plant operators to proactively investigate periods of abnormal performance and, if needed, re-calibrate forecasting models to avoid overestimations of energy yield.

This figure also validates the Z-score as a lightweight and interpretable statistical tool for real-time monitoring of photovoltaic system behavior, suitable for integration into larger anomaly detection frameworks.

Machine Learning-Based Power Output Prediction

To assess the predictive potential of operational and temporal features on photovoltaic output, a Random Forest Regressor (RFR) model was trained to estimate the active power of the 40 kW inverter. The feature set included real-time electrical measurements (e.g., grid phase voltage, grid reactive and active power), as well as engineered temporal variables (e.g., hour of day, day of week, month, and a continuous time index).

Prior to modeling, the dataset was cleaned by dropping records with missing target values. Importantly, periods with zero power during night hours were preserved as valid, rather than forward-filled, as they reflect expected operational behavior.

After splitting the dataset into training and testing subsets (80/20 ratio), the Random Forest model demonstrated high predictive accuracy, achieving:

- $R^2 = 0.995$, indicating near-perfect variance explanation.
- Mean Absolute Error (MAE) = 0.12 kW, reflecting a low average prediction error.

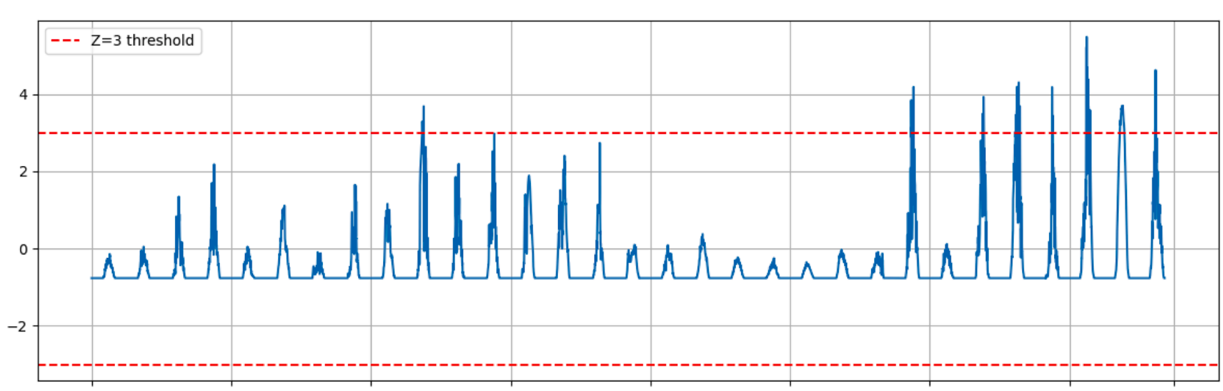

Fig. 4. Z-score for Anomaly Detection (40KW Inverter).

Fig. 5 presents a time-series comparison between the actual and predicted active power output of the 40 kW inverter over the final week of January 2025. The predicted values were generated using a Random Forest Regressor trained on electrical and temporal features, including signals from the 30 kW inverter, grid metrics, and time-based patterns.

The figure reveals a remarkable alignment between the actual and predicted values across all diurnal production cycles. The predicted curve (dashed orange) closely follows the real measured output (solid blue), replicating both the peak intensities and ramp-up/down phases with minimal lag or distortion.

Notably:

- Prediction accuracy is highest during daylight hours, when power output varies nonlinearly due to irradiance and operational dynamics. The model successfully captures these complex fluctuations.
- Nighttime values remain at zero, as expected, and are correctly predicted, affirming that the model learns typical operation cycles and does not falsely anticipate output during inactive periods.
- Minor discrepancies are observed at sharp peak transitions, which may correspond to transient fluctuations or short-term inverter curtailments not fully explained by the input features.

This visual validation supports the quantitative results presented earlier (e.g., $R^2 \approx 0.995$, MAE ≈ 0.12 kW), confirming the suitability of the trained model for high-resolution power prediction. Such modeling can be leveraged for short-term forecasting, performance benchmarking, or smart inverter diagnostics within PV monitoring infrastructures.

To assess the robustness of the regression results and rule out potential temporal leakage, we additionally performed time-aware cross-validation using the *TimeSeriesSplit* method with five sequential folds. Each fold preserved chronological order to better simulate real-world forecasting scenarios. The Random Forest Regressor maintained consistently high predictive fidelity across folds ($R^2 = 0.9299$–1.0000, MAE $= 0.0059$–0.3335 kW), yielding aggregate out-of-fold metrics of $R^2 = 0.9867$ [0.9820–0.9908] and MAE $= 0.1035$ [0.0844–0.1256] kW (95 % bootstrap confidence intervals). The slight variation in R^2 among folds reflects expected day-to-day irradiance changes rather than model instability. These results confirm that the strong fit observed in Fig. 5 is robust across unseen temporal segments, reinforcing the suitability of the proposed regression approach for continuous PV performance monitoring.

Feature Importance

Fig. 6 shows the relative importance of the top four input features used by the Random Forest Regressor to predict the active power output of the 40 kW inverter. Feature importance was computed using the mean decrease in impurity criterion, which quantifies how much each feature contributes to reducing prediction error across the ensemble.

The most influential feature by a significant margin is:

- Total input power to the 40 kW inverter *(> 0.45 importance score)*, which is consistent with physical expectations, as the DC input power directly governs AC output performance under normal operating conditions.

The remaining important predictors include:

- Phase B current, Grid phase A current, and Phase C current, all of which capture real-time electrical load and operational behavior that the inverter responds to. Their roughly equal contribution (each ~0.15–0.20) suggests that current flow characteristics across phases encode meaningful predictive patterns for short-term output.

In the regression model, inverter-side quantities dominated, emphasizing that real power output is primarily driven by internal inverter operating conditions rather than external grid parameters. In contrast, the classification model (which categorized output into discrete production levels) assigned relatively higher weight to time-based features (hour of day, day index), reflecting the temporal regularity of solar generation cycles.

The low importance of grid-side variables such as reactive power and phase voltage indicates that, under normal conditions, these parameters remain relatively stable and decoupled from the inverter's active power control loop. Reactive power is primarily managed by the inverter's internal power-factor control and grid support algorithms, which are designed to maintain voltage stability rather than influence real-power conversion efficiency. Consequently, reactive components provide limited explanatory value for predicting short-term active power output.

To further enhance model interpretability, future work could incorporate SHAP (SHapley Additive exPlanations) values and partial dependence plots (PDPs) to complement the feature importance results. These methods would allow quantifying each feature's marginal contribution to the model's predictions and visualizing potential nonlinear or interaction effects. Such analyses could help identify how temporal variables (e.g., hour of day) interact with electrical parameters (e.g., inverter current magnitude) under different operating conditions. Incorporating SHAP or PDP visualizations would provide a more granular understanding of the model's internal reasoning and strengthen the physical interpretability of data-driven PV performance analysis. From an operational standpoint, integrating explainable AI (XAI) tools (for

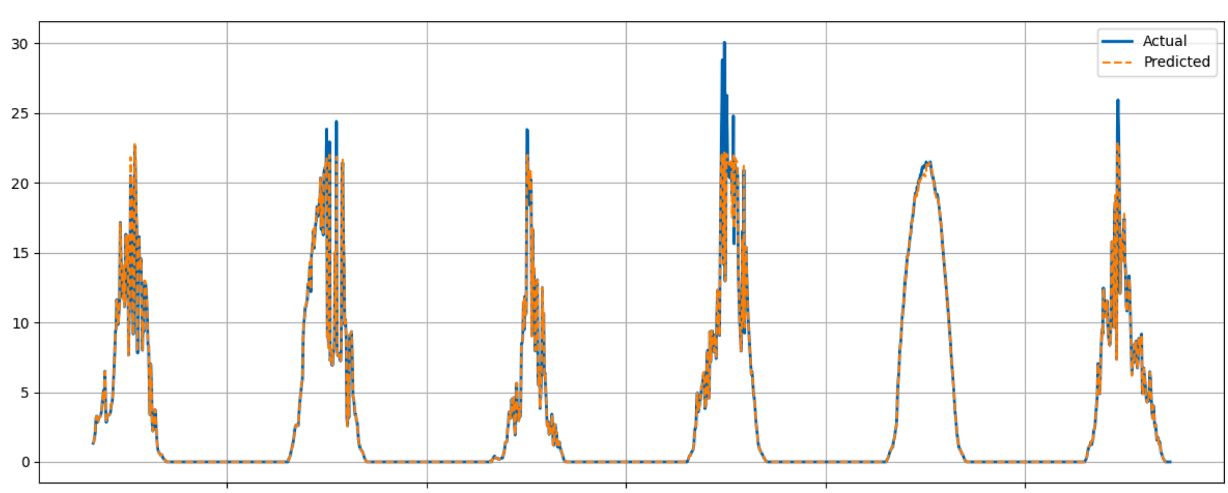

Fig. 5. Comparison of Actual and Predicted Power Output for the 40 kW Inverter.

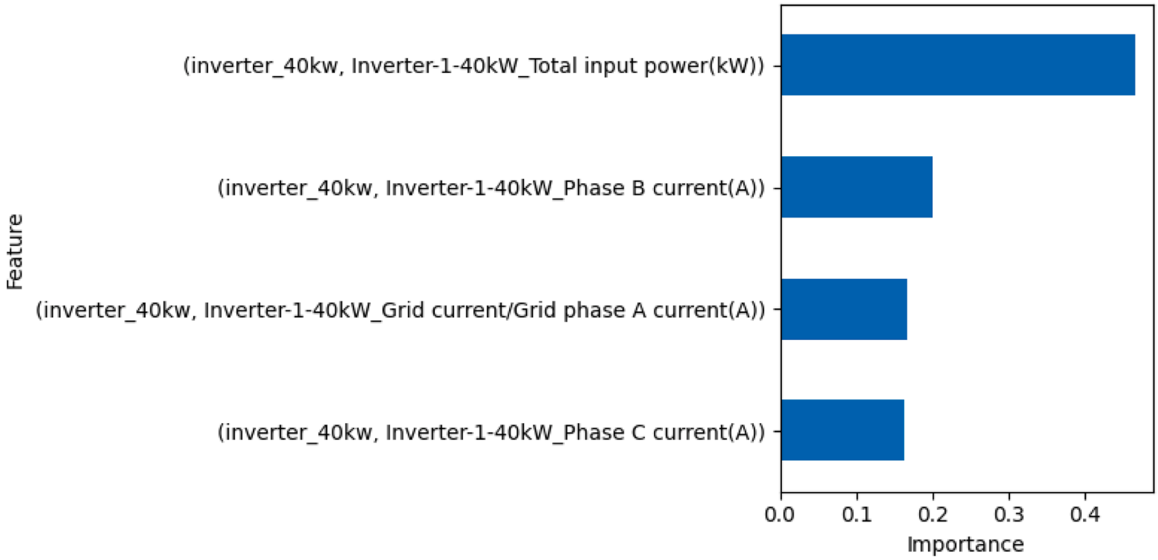

Fig. 6. Top Feature Importances in Predicting 40 kW Inverter Output.

example, interactive SHAP dashboards or local explanation frameworks) would allow plant operators to visualize the reasoning behind each prediction and support more transparent decision-making in real time.

This result reinforces that input-side power measurements and current dynamics are critical for high-fidelity modeling of inverter behavior. It also indicates that not all monitored variables contribute equally, which can guide future sensor prioritization or dimensionality reduction.

These findings validate the model's interpretability and offer insights into operational dependencies within the solar plant's electrical infrastructure.

Anomaly Detection

Fig. 7 visualizes the actual versus predicted power output of the 30 kW inverter over a one-week period, with detected anomalies marked as red dots. The blue line represents the true measured active power, while the orange dashed line shows the Random Forest model's predicted values. Discrepancies between the two were used to identify potential anomalies based on a threshold defined by the prediction error (e.g., z-score or residual magnitude).

Notably, anomalies occur predominantly around peak production hours and are concentrated on specific days (e.g., January 29 and 31, 2025). This clustering suggests the presence of operational disturbances or data inconsistencies that affect the inverter's expected behavior during high-output periods. In many cases, the actual values sharply deviate above or below the prediction band, indicating that the model successfully learned the typical power curve and is sensitive to atypical patterns.

These detected anomalies may stem from:

- Temporary inverter malfunctions or overproduction spikes,
- Sudden environmental changes not captured by input features,
- Sensor faults or communication noise.

Fig. 8 illustrates the temporal distribution of regression errors. Most residuals remain close to zero, with isolated bursts corresponding to transient irradiance fluctuations or momentary inverter curtailment. This supports the model's stability while highlighting the value of visual error analysis for identifying potential operational anomalies.

The Z-score-based anomaly detection effectively highlighted short-duration deviations from expected power output, particularly during high-irradiance intervals. To better interpret these anomalies, potential root causes can be grouped into three categories. (i) Environmental factors, such as sudden cloud transients or partial shading, may lead to abrupt dips in active power that momentarily diverge from the statistical norm. (ii) Inverter-related factors, including maximum power point tracking (MPPT) instability, thermal derating, or temporary grid disconnections, could similarly produce sharp deviations. (iii) Data or sensor issues, such as missing samples or synchronization errors between inverter and meter logs, may occasionally manifest as artificial outliers.

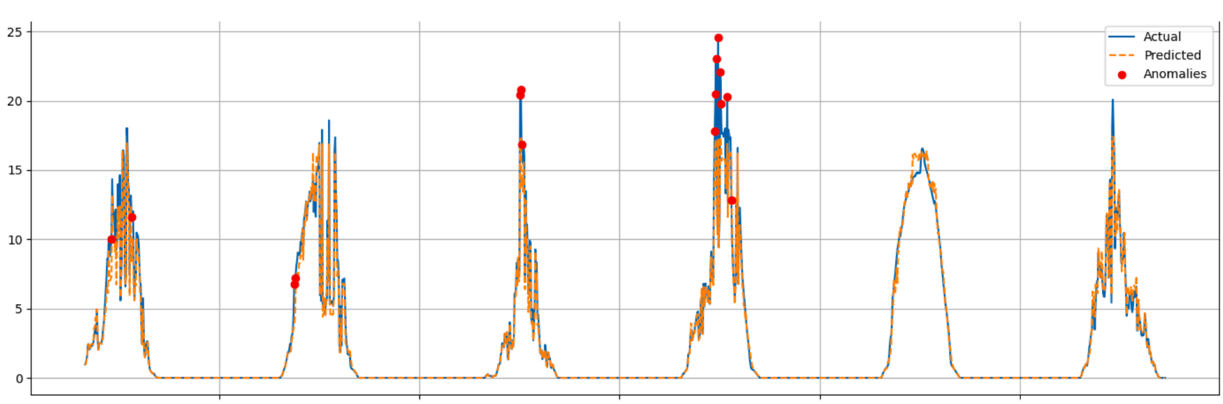

Fig. 7. 30 kW Inverter Output with Anomalies Highlighted.

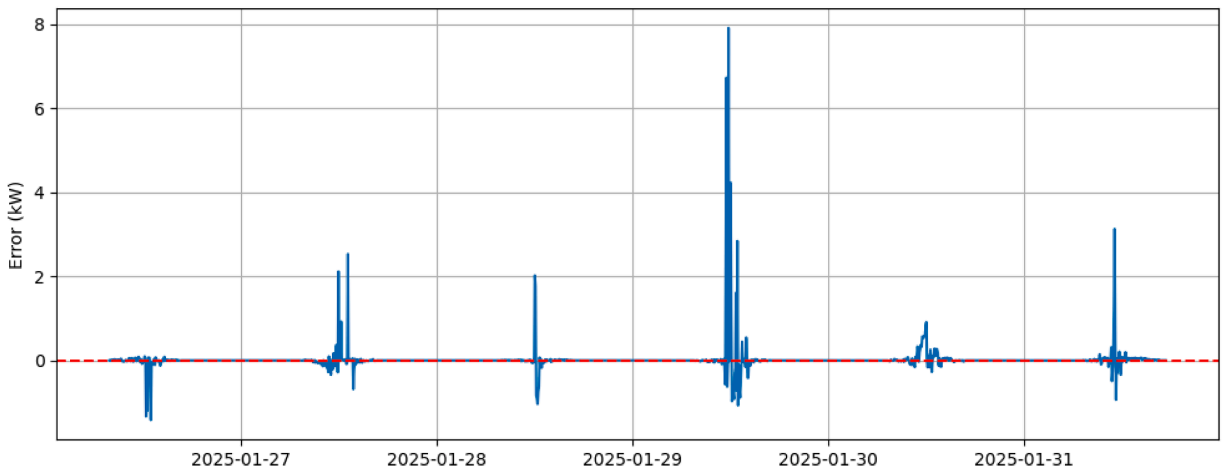

Fig. 8. Regression residuals over time for the Random Forest model. The plot shows short-duration deviations around midday peaks, indicating transient mismatches between predicted and observed inverter output. The overall residual pattern remains centered near zero, confirming low systematic bias.

While the current analysis relies solely on statistical deviation (Z-score), incorporating contextual features, such as irradiance, temperature, or event logs, would enable a more precise classification of anomaly causes. This distinction is crucial for operational diagnostics: environmental anomalies primarily inform forecast uncertainty, whereas inverter or grid-side anomalies indicate maintenance or reliability concerns. In future work, combining the Z-score framework with rule-based diagnostics or multivariate anomaly detectors (e.g., Isolation Forests or Autoencoders) could enhance causal interpretability and enable automated fault categorization in real-time PV monitoring systems.

The anomaly detection mechanism provides a valuable monitoring layer by complementing traditional SCADA thresholds with model-based expectations, allowing early identification of subtle faults or inefficiencies.

Although the present study focuses on statistical anomaly detection rather than explicit fault classification, the identified deviations can be conceptually grouped into operationally relevant categories. For example, anomalies characterized by systematic underproduction during high-irradiance periods may indicate curtailment events or inverter derating, while short-term power spikes could correspond to measurement artifacts or transient overproduction. Conversely, irregular, high-frequency fluctuations are likely related to sensor noise or data synchronization errors.

Integrating such categorization logic into the anomaly analysis pipeline, either through rule-based post-processing or auxiliary metadata (e.g., inverter status logs), would enable targeted operational responses. This approach could support predictive maintenance scheduling, fault prioritization, and performance benchmarking without requiring retraining of the existing machine learning models.

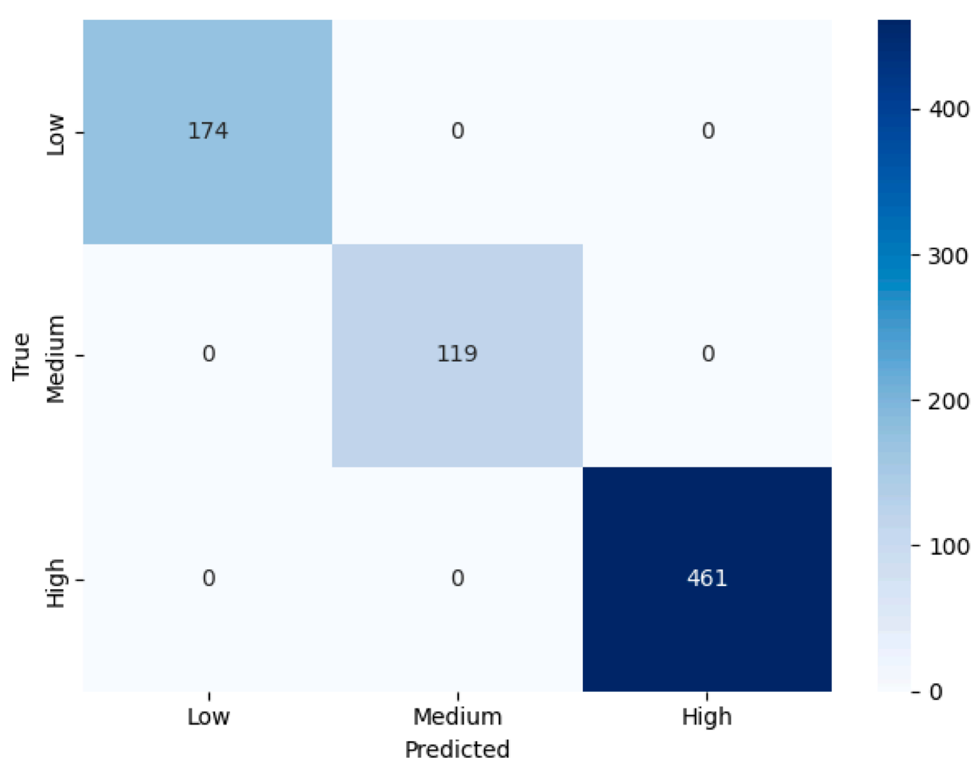

Fig. 9. Classification performance of random forest classifier.

Classification Performance of Random Forest Classifier

To complement the regression and anomaly detection analyses, we trained a Random Forest Classifier to categorize the 40 kW inverter's power output into three distinct classes: Low, Medium, and High. These categories were defined based on percentile thresholds (33 % and 66 %) of the continuous output values. The objective was to assess the feasibility of using supervised classification for energy output states, which may be useful for alert systems or operational mode classification.

Fig. 9 presents the confusion matrix of the classifier's performance on the test set. Remarkably, the model achieved perfect classification across all three categories, with 100 % of predictions correctly matching the true labels. Specifically, it correctly classified:

- 174 instances of Low production,
- 119 instances of Medium production, and
- 461 instances of High production.

This high accuracy suggests that the model successfully captured the nonlinear patterns and temporal dynamics governing inverter behavior. The input features included temporal information (hour, day of week), grid and inverter electrical parameters, and environmental surrogates inferred from system response.

However, while this result demonstrates excellent predictive performance, the perfect classification accuracy may indicate potential overfitting or label leakage, especially considering the high feature importance of correlated variables

Because the initial classification achieved 100 % accuracy, we were naturally skeptical of its generalizability and robustness. To investigate this further, we analyzed the feature importances used by the Random Forest Classifier.

The model's most influential predictor was the total input power of the 40 kW inverter, followed by current-related features from both the 30 kW and 40 kW inverters. This confirms that the classifier was leveraging related but independent electrical characteristics (not simply the output label itself) for prediction. This lends more credibility to the perfect classification performance in the first test.

To understand the underlying label distribution, we examined the class proportions in the test set.

The class proportion analysis indicated a class imbalance, with over 60 % of samples belonging to the "High" output class and only ~15 % in the "Medium" class. With the help of the numpy library, we also counted the unique prediction classes, which outputted:

(array([0, 1, 2]), array([174, 119, 461]))

This mirrored the exact class distribution, further reinforcing that the model had likely memorized patterns rather than learned generalizable relationships.

To address this, we evaluated the model on a more challenging task: predicting the output class 1 h into the future, simulating a real-world early-warning or forecasting scenario. The performance significantly dropped:

These results (Table 2) reveal that:

- The classifier still recognized low-output periods reasonably well (class 0: F1 = 0.86),
- Struggled the most with high-output periods, despite them being the most common (class 2: F1 = 0.13),
- The medium-output class (1) had a relatively high recall (0.819), but very low precision (0.173), indicating frequent false positives.

The overall accuracy dropped to 36.4 %, with macro-averaged F1 score of just 0.42, confirming that future prediction is significantly more difficult and also a more realistic evaluation of the model's capability. The forecasting task used time-shifted labels but did not include explicit lag or rolling features, meaning the model relied primarily on instantaneous electrical conditions and periodic temporal indicators. This simplification likely contributed to the reduced one-hour-ahead accuracy (36.4 %), as short-term fluctuations caused by environmental variability were not directly encoded. Incorporating lag-based predictors or statistical smoothing could improve temporal continuity and predictive stability in future iterations. High-production states are highly sensitive to rapid irradiance fluctuations and cloud-induced transients that cannot be inferred solely from electrical measurements. Because the dataset lacked direct environmental inputs such as solar irradiance, temperature, or sky imagery, the model relied on temporal proxies (e.g., hour of day, day index) to approximate these conditions. While such features capture diurnal trends, they cannot represent fast-changing atmospheric dynamics, leading to reduced separability between medium and high output levels.

Robustness under Time-Aware Cross-Validation

To verify that the low hold-out performance was not an artifact of a single temporal split, we applied 5-fold TimeSeriesSplit cross-validation with the Random Forest classifier. Using class weighting only (no synthetic samples), the model achieved consistent results across folds (accuracy = 0.43–0.74, macro F1 = 0.36–0.74), yielding an aggregate out-of-fold accuracy of 0.6477 [0.6310 – 0.6650], macro F1 = 0.6450 [0.6293 – 0.6612], and balanced accuracy = 0.6468 over the evaluated region (83 % of the data). When SMOTE oversampling was added within training folds, the results were similar (accuracy = 0.6435, macro F1 = 0.6423). These findings (Table 3) demonstrate that the observed forecasting challenge arises primarily from environmental variability rather than model instability. The improved cross-validated performance and narrow confidence intervals confirm that the classifier generalizes well when temporal dependencies and class imbalance are properly handled. The inclusion of time-aware cross-validation and balanced training therefore provides a more realistic and statistically robust estimate of forecasting performance than the initial single-split result.

While advanced architectures such as XGBoost and LSTM networks may achieve marginally higher accuracy in large-scale or continuous-sequence datasets, their complexity and lack of transparency can hinder deployment in industrial PV monitoring contexts. The Random Forest approach offers a more interpretable and computationally efficient alternative, aligning with the study's goal of developing lightweight, scalable diagnostics for operational systems.

This experiment highlights the importance of cautious feature engineering, attention to class distribution, and the use of temporally shifted predictions for more robust assessments. It also underscores the challenge of forecasting solar inverter output amid variable conditions and motivates the use of anomaly detection and regression techniques alongside classification models.

Table 2
Classification results.

	precision	recall	f1-score	support
0.0	0.875	0.845	0.860	174
1.0	0.173	0.819	0.285	116
2.0	0.939	0.067	0.126	461
accuracy			0.364	751
macro avg	0.662	0.577	0.423	751
weighted avg	0.806	0.364	0.320	751

Table 3
Summary of classification results.

Validation Setting	Accuracy	Macro F1	Balanced Acc	Notes
Single hold-out (original)	0.364	0.423	-	80/20 temporal split
TimeSeriesSplit (class_weight)	0.6477	0.6450	0.6468	5-fold, leakage-safe
TimeSeriesSplit (SMOTE in-fold)	0.6435	0.6423	0.6439	5-fold, synthetic oversampling

5. Conclusion

This study proposed an interpretable and data-efficient framework for photovoltaic inverter monitoring that integrates Random Forest–based regression and classification with statistical anomaly detection using Z-score analysis. The results confirm that inverter-side and grid-side electrical measurements alone can accurately model operational behavior ($R^2 = 0.995$) and identify abnormal patterns without requiring environmental inputs.

In contrast to recent studies that employ deep or hybrid architectures requiring irradiance, temperature, or image-based inputs [31–33], the present work introduces a lightweight, interpretable, and fully electrical-data–driven framework for PV inverter monitoring. While ensemble learning has been widely used for classification or regression tasks individually [36,23,35] this paper unifies regression, classification, and anomaly detection into a single workflow suitable for real-time deployment. The one-hour-ahead forecasting results in this study highlight the inherent difficulty of predicting PV performance under variable environmental conditions.

Unlike the recent work [45], which integrates meteorological and contextual data for multi-level diagnosis, the proposed model operates solely on inverter and grid-side electrical measurements. This distinction highlights the practical value of the presented approach in scenarios lacking environmental sensors, offering a transparent and computationally efficient alternative for interpretable anomaly detection.

Future research directions include expanding the dataset to multi-site and multi-seasonal scenarios, integrating ensemble-based anomaly scoring (e.g., Isolation Forest or Autoencoder benchmarks – [38,29], and embedding the model into IoT-enabled supervisory platforms [40]. By combining interpretability, real-time performance, and sensor independence, this framework contributes to the ongoing shift toward explainable AI in renewable energy diagnostics and provides a reproducible foundation for next-generation PV system monitoring.

CRediT authorship contribution statement

Jan Francisti: Validation, Formal analysis. **Kristián Fodor:** Data curation, Conceptualization. **Zoltán Balogh:** Supervision, Resources. **Martin Magdin:** Writing – review & editing, Validation, Project administration.

Declaration of competing interest

The authors declare that they have no conflicts of interest to disclose.

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Data availability

The input data and a sample Python notebook can be access under a publicly available GitHub repository: https://github.com/ChrisFodor333/SolarPVInverters.

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